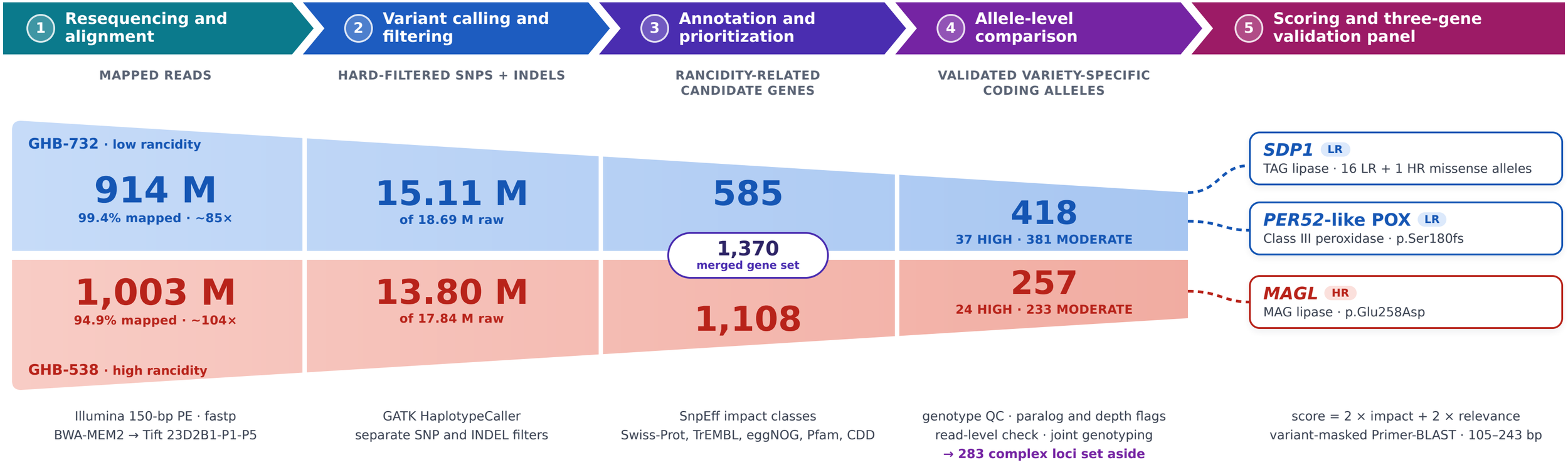

1 Resequencing and alignment
2 Variant calling and filtering
3 Annotation and prioritization
4 Allele-level comparison
5 Scoring and three-gene validation panel
MAPPED READS
HARD-FILTERED SNPS + INDELS
RANCIDITY-RELATED CANDIDATE GENES
VALIDATED VARIETY-SPECIFIC CODING ALLELES
GHB-732 · low rancidity
914 M
99.4% mapped · ~85×
15.11 M
of 18.69 M raw
585
418
37 HIGH · 381 MODERATE
1,370
merged gene set
1,003 M
94.9% mapped · ~104×
13.80 M
of 17.84 M raw
1,108
257
24 HIGH · 233 MODERATE
GHB-538 · high rancidity
SDP1 LR
TAG lipase · 16 LR + 1 HR missense alleles
PER52-like POX LR
Class III peroxidase · p.Ser180fs
MAGL HR
MAG lipase · p.Glu258Asp
Illumina 150-bp PE · fastp
BWA-MEM2 → Tift 23D2B1-P1-P5
GATK HaplotypeCaller
separate SNP and INDEL filters
SnpEff impact classes
Swiss-Prot, TrEMBL, eggNOG, Pfam, CDD
genotype QC · paralog and depth flags
read-level check · joint genotyping
→ 283 complex loci set aside
score = 2 × impact + 2 × relevance
variant-masked Primer-BLAST · 105–243 bp